

Enhanced Entity-Relationship (EER) Diagram Walkthrough

System Design for Retail Chain Management

Course: Database Management System

Course Code: CSA0503

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Introduction & Business Scope

- System Overview:
 - A Retail Chain Management System handles multi-location operations, inventory, sales, employees and suppliers.
- Core Functional Domains:
 - • Store & Warehouse Network
 - • Product Catalog
 - • Staff Management
 - • Sales & Customer Loyalty

Core Entities & Attributes

- Store: StoreID, StoreName, Address, Phone
- Product: ProductID, SKU, Name, UnitPrice, StockQty
- Employee: EmpID, Name, HireDate, Role, Salary
- Customer: CustomerID, Name, Phone, Email, Points
- Supplier: SupplierID, CompanyName, ContactNo
- Transaction: TxnID, TxnDate, TotalAmount, PaymentMethod

Advanced EER Constructs & Concepts

- Specialization:
- Employee -> StoreAssociate, Manager
- Product -> PerishableItem, NonPerishableItem

- Aggregation:
- Product SUPPLIED_BY Supplier aggregated as SupplyContract linked with Store.

Relationships & Structural Constraints

- EER Architecture:
- Manager -> Store
- Employee <-> Transaction
- Transaction <-> Product

- Cardinality:
- Store-Employee (1:N)
- Store-Product (N:M via StoreInventory)
- Transaction-Product (N:M via TransactionLineItem)

Relational Mapping & Conclusion

- Store(StoreID,...)
- Employee(EmpID,...,StoreID,ManagerID)
- Product(ProductID,...)
- PerishableItem(ProductID,...)
- StoreInventory(StoreID,ProductID,...)
- Transaction(TxnID,...)

- Conclusion:
- Data integrity, scalability and efficient retail management.